EXPERIMENT-5
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Objective :-1. WAP to train and evaluate a convolutional neural network using Keras Library to classify MNIST fashion dataset. Demonstrate the effect of filter size, regularization, batch size and optimization algorithm on model performance.

Description :-

- Implemented a Convolutional Neural Network (CNN) using Keras with TensorFlow for image classification.
- Used the Fashion-MNIST dataset containing 28×28 grayscale images of 10 fashion categories.
- Preprocessed the dataset by normalizing pixel values and reshaping images for CNN input.
- Designed a CNN architecture with convolution, pooling, flatten, dense, and softmax layers.
- Trained the model for multiple epochs and evaluated performance using accuracy and loss metrics.
- Plotted training and validation graphs to analyze model learning behavior.
- Conducted experiments to study the effect of filter size, regularization, batch size, and optimizer on model performance.

Source Code :-

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

from tensorflow.keras.datasets import fashion_mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense,
Dropout
from tensorflow.keras.optimizers import Adam, SGD, RMSprop
from tensorflow.keras.regularizers import l2
```

```
(x_train, y_train), (x_test, y_test) = fashion_mnist.load_data()

# Normalize pixel values
x_train = x_train / 255.0
x_test = x_test / 255.0

# Reshape data for CNN
x_train = x_train.reshape(-1,28,28,1)
x_test = x_test.reshape(-1,28,28,1)

print("Training Data Shape:", x_train.shape)
print("Testing Data Shape:", x_test.shape)
Training Data Shape: (60000, 28, 28, 1)
Testing Data Shape: (10000, 28, 28, 1)
def create_model(filter_size=3, regularization=False, optimizer='adam'):

    if regularization:
        reg = l2(0.001)
    else:
        reg = None

    model = Sequential()

    model.add(Conv2D(32,(filter_size,filter_size),
                     activation='relu',
                     kernel_regularizer=reg,
                     input_shape=(28,28,1)))

    model.add(MaxPooling2D((2,2)))

    model.add(Conv2D(64,(filter_size,filter_size),activation='relu'))

    model.add(MaxPooling2D((2,2)))

    model.add(Flatten())

    model.add(Dense(128,activation='relu'))

    model.add(Dropout(0.3))
```

```
    model.add(Dense(10,activation='softmax'))

    if optimizer == 'adam':
        opt = Adam()
    elif optimizer == 'sgd':
        opt = SGD()
    else:
        opt = RMSprop()

    model.compile(
        optimizer=opt,
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )

    return model
model = create_model()

history = model.fit(
        x_train,
        y_train,
        epochs=10,
        batch_size=64,
        validation_split=0.2,
        verbose=1
)
test_loss, test_acc = model.evaluate(x_test,y_test)

print("\nTest Accuracy:",test_acc)
print("Test Loss:",test_loss)
```

Output :-

```
Test Accuracy: 0.9089999794960022
Test Loss: 0.254584938287735
```

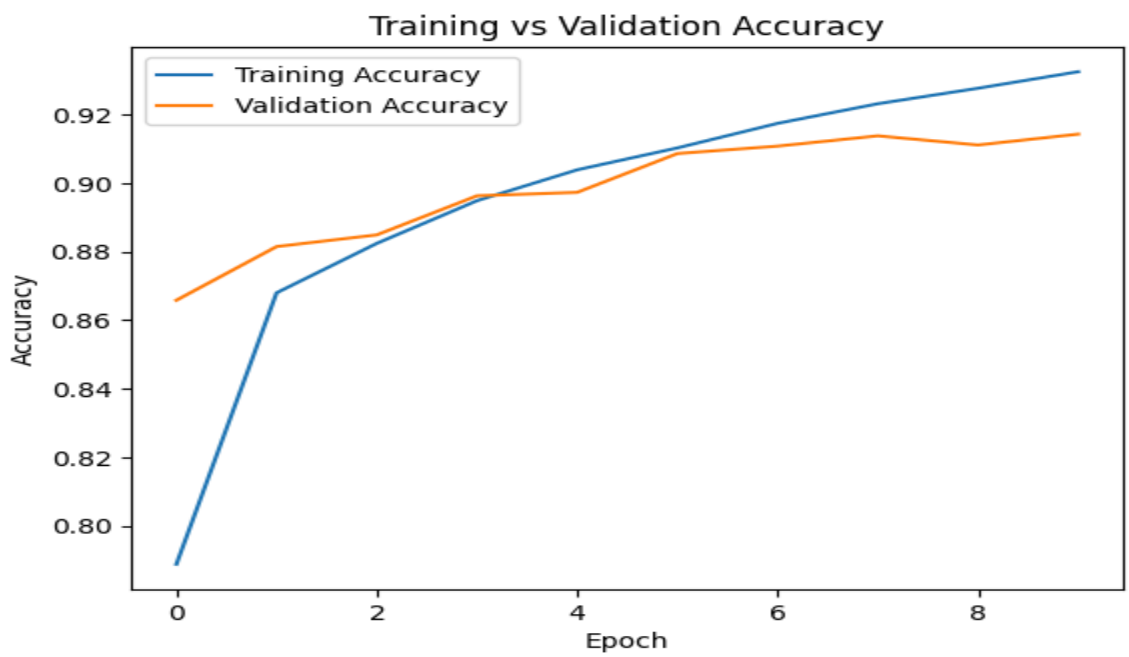
```
epoch_table = pd.DataFrame(history.history)
print("\nEpoch-wise Performance\n")
print(epoch_table)
Epoch-wise Performance
```

Output :-

```
   accuracy      loss  val_accuracy  val_loss
0  0.788750  0.571954      0.865833  0.371731
1  0.868000  0.366708      0.881500  0.325045
2  0.882438  0.319284      0.884917  0.307075
3  0.894896  0.287117      0.896333  0.282052
4  0.903917  0.262578      0.897333  0.273392
5  0.910312  0.241184      0.908667  0.248109
6  0.917479  0.224634      0.910833  0.240539
7  0.923208  0.206786      0.913833  0.237461
8  0.927729  0.192864      0.911167  0.250773
9  0.932562  0.180979      0.914333  0.240931
```

```
plt.figure()

plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])

plt.title("Training vs Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend(["Training Accuracy","Validation Accuracy"])
plt.show()
```

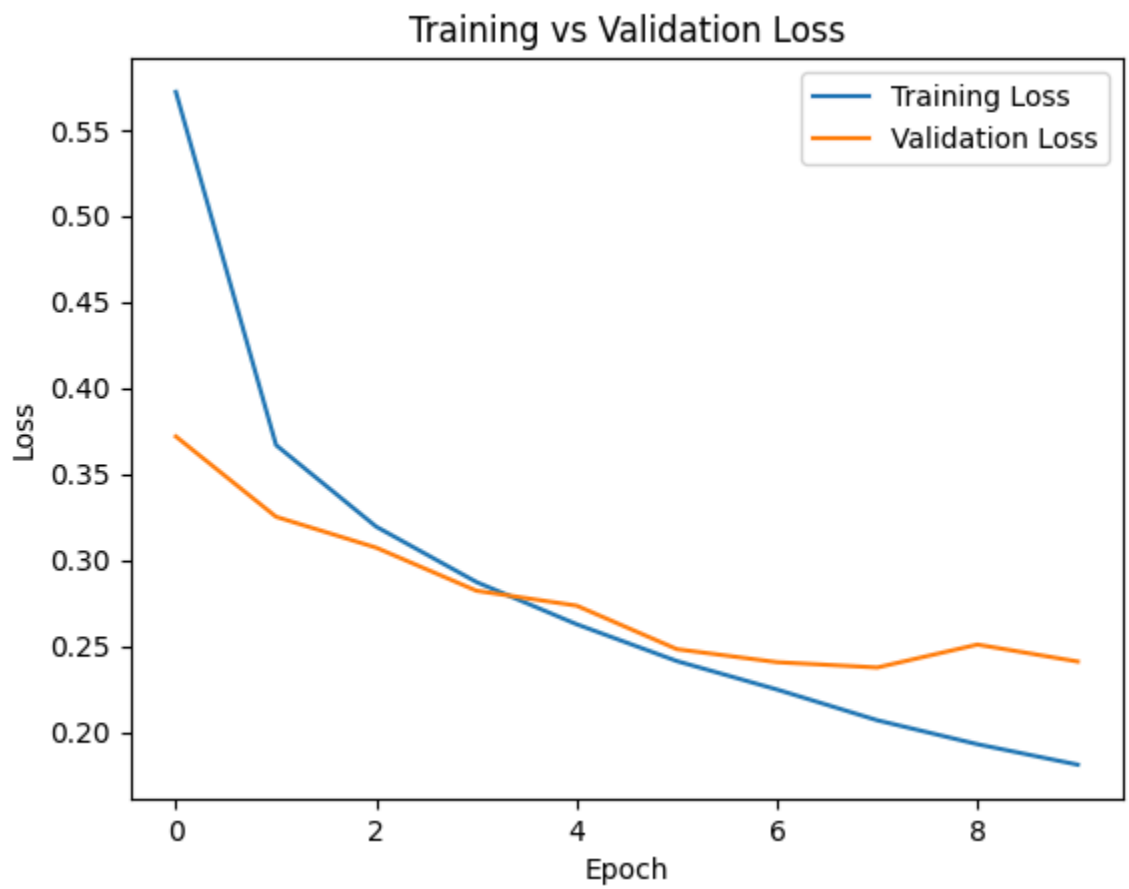
Output :-




```
plt.figure()
```

```
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])

plt.title("Training vs Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend(["Training Loss","Validation Loss"])

plt.show()
```

Output :-

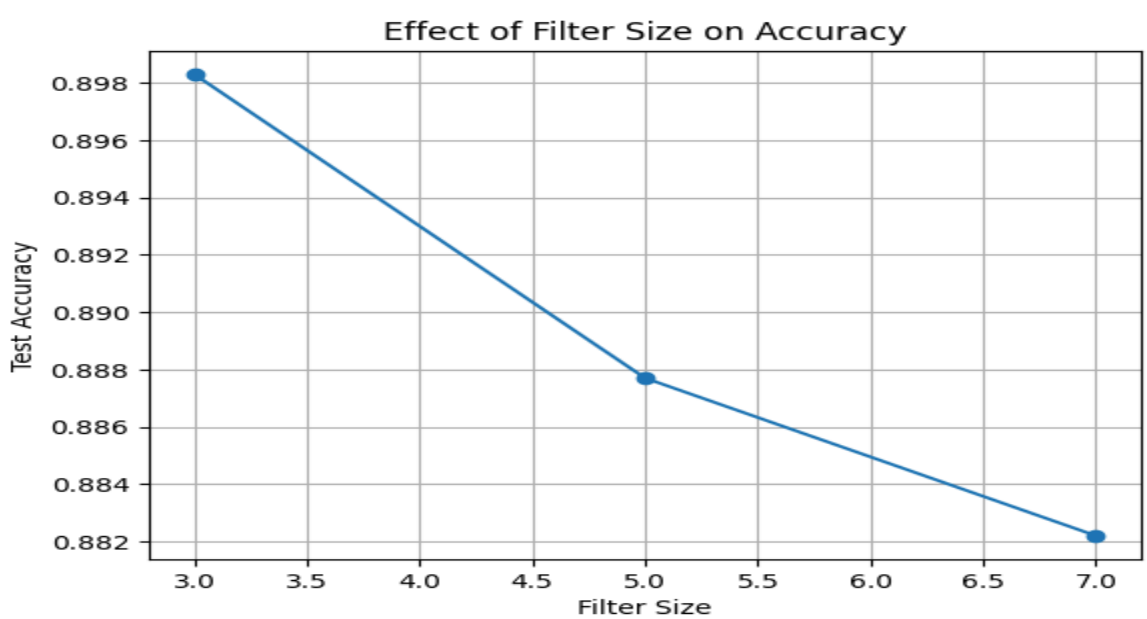


EXPERIMENT 1 : Effect of Filter Size

```
filter_sizes = [3,5,7]
filter_accuracy = []
for f in filter_sizes:
```

```
    print("\nTraining with filter size:",f)
    model = create_model(filter_size=f)
    model.fit(
        x_train,
        y_train,
        epochs=5,
        batch_size=64,
        validation_split=0.2,
        verbose=1
    )

    loss,acc = model.evaluate(x_test,y_test)

    filter_accuracy.append(acc)

plt.figure()

plt.plot(filter_sizes,filter_accuracy,marker='o')

plt.title("Effect of Filter Size on Accuracy")
plt.xlabel("Filter Size")
plt.ylabel("Test Accuracy")
plt.grid(True)
plt.show()
```

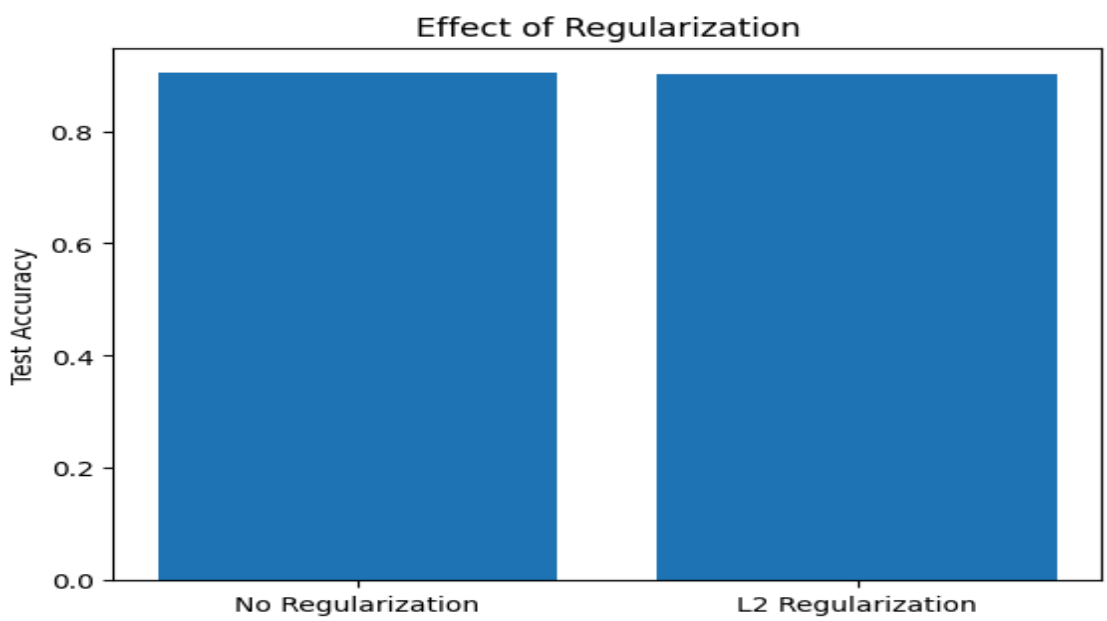
Output :-



EXPERIMENT 2 : Effect of Regularization

```
reg_types = [False,True]
reg_accuracy = []
for r in reg_types:

    print("\nTraining with Regularization:",r)
    model = create_model(regularization=r)
    model.fit(
        x_train,
        y_train,
        epochs=5,
        batch_size=64,
        validation_split=0.2,
        verbose=1
    )
    loss,acc = model.evaluate(x_test,y_test)
    reg_accuracy.append(acc)

plt.figure()
labels = ["No Regularization","L2 Regularization"]
plt.bar(labels,reg_accuracy)
plt.title("Effect of Regularization")
plt.ylabel("Test Accuracy")
plt.show()
```

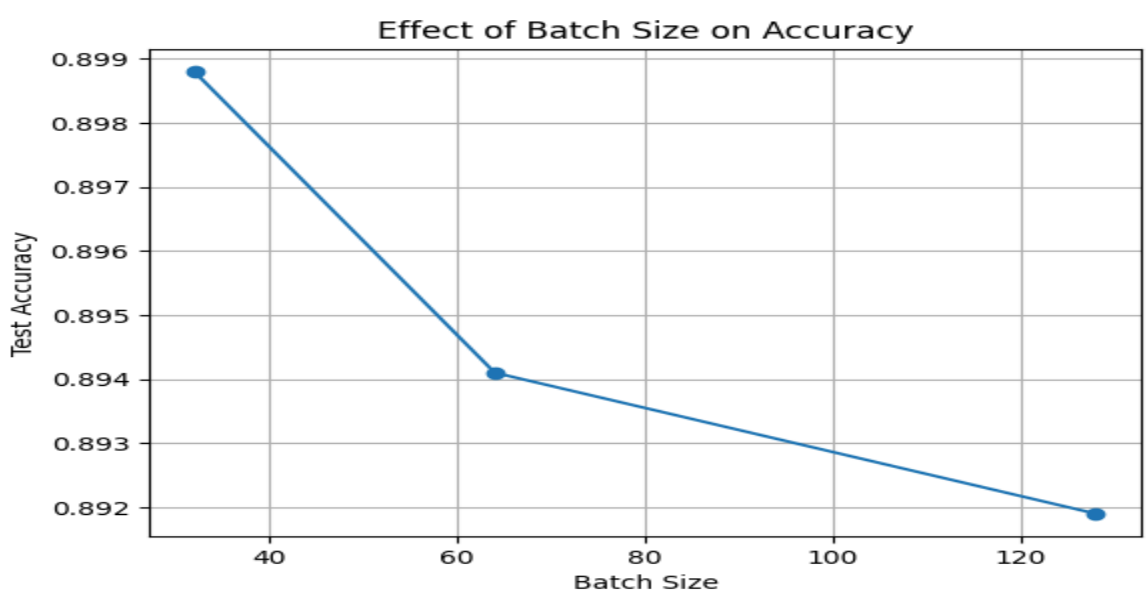
Output :-



EXPERIMENT 3 : Effect of Batch Size

```
batch_sizes = [32,64,128]
batch_accuracy = []
for b in batch_sizes:
    print("\nTraining with batch size:",b)
    model = create_model()
    model.fit(
        x_train,
        y_train,
        epochs=5,
        batch_size=b,
        validation_split=0.2,
        verbose=1 )
    loss,acc = model.evaluate(x_test,y_test)
    batch_accuracy.append(acc)

plt.figure()
plt.plot(batch_sizes,batch_accuracy,marker='o')
plt.title("Effect of Batch Size on Accuracy")
plt.xlabel("Batch Size")
plt.ylabel("Test Accuracy")
plt.grid(True)
plt.show()
```

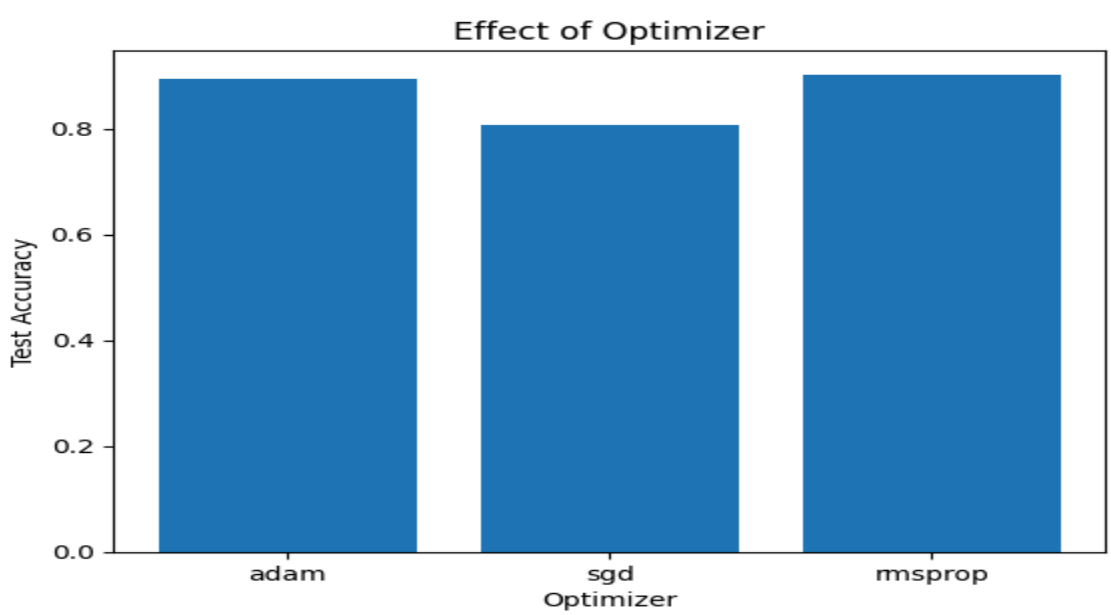
Output :-



EXPERIMENT 4 : Effect of Optimizer

```
optimizers = ['adam','sgd','rmsprop']
opt_accuracy = []
for opt in optimizers:
    print("\nTraining with optimizer:",opt)
    model = create_model(optimizer=opt)
    model.fit(
        x_train,
        y_train,
        epochs=5,
        batch_size=64,
        validation_split=0.2,
        verbose=1
    )
    loss,acc = model.evaluate(x_test,y_test)
    opt_accuracy.append(acc)

plt.figure()
plt.bar(optimizers,opt_accuracy)
plt.title("Effect of Optimizer")
plt.xlabel("Optimizer")
plt.ylabel("Test Accuracy")
plt.show()
```

Output :-



MY COMMENTS :-

From this experiment, I learned how to design and train a CNN model for image classification tasks. We also understood the importance of preprocessing image data, selecting appropriate model architecture, and tuning hyperparameters. The experiment demonstrated that parameters such as filter size, regularization, batch size, and optimizer significantly influence the performance and efficiency of deep learning models.

Learning from Variation Experiments

1. Effect of Filter Size
Different filter sizes such as 3×3, 5×5, and 7×7 were tested. It was observed that smaller filters like 3×3 generally produced better performance because they capture fine-grained features and require fewer parameters.

2. Effect of Regularization
L2 regularization was applied to reduce overfitting. It was observed that regularization helps improve model generalization by preventing the weights from becoming excessively large.

3. Effect of Batch Size
Different batch sizes such as 32, 64, and 128 were experimented with. Smaller batch sizes provided more stable learning, while larger batch sizes reduced training time but sometimes slightly reduced accuracy.

4. Effect of Optimization Algorithm
Different optimizers such as Adam, SGD, and RMSprop were tested. Among them, Adam generally converged faster and provided higher accuracy compared to other optimizers.